

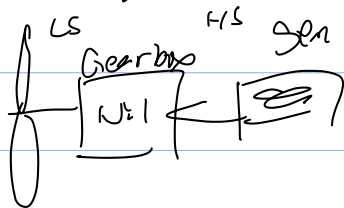
Induction generator.

$$P_{mech} = \frac{1}{2} \rho A R^2 C_p V^3$$

V : wind speed

$$\rho = 1.225 \text{ kg/m}^3$$

$$T_{SR} = \frac{WR}{V}$$



$$P_{HS} = \frac{1}{2} \rho A R^2 C_p V^3$$

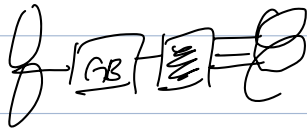
$$P_{HS} = P_{LS}$$

$$T_{LS} = \frac{1}{2} \rho A R^2 C_t V^3$$

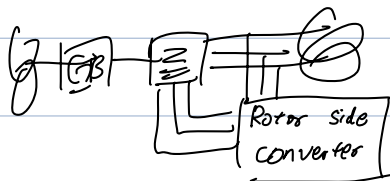
$$T_{HS} = \frac{1}{N} T_{LS}$$

$$\omega_{HS} = N \omega_{LS}$$

Induction Generator Topologies:



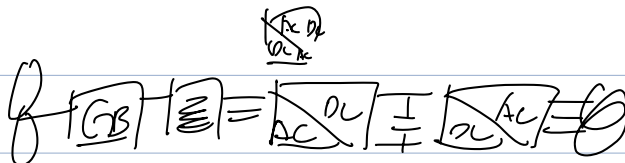
Fixed speed. IG.



Doubly Fed Induction Generator

DFIG

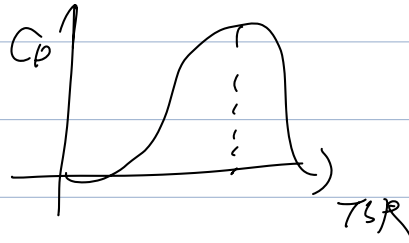
Limited speed range.



Full converter IG

Variable speed.

Variable speed IG:



$$TSR = \frac{\omega R}{V} = \text{constante}$$

$$\Rightarrow \frac{\omega}{V} = \text{const.}$$

$$\Rightarrow \frac{N}{V} = \text{const}$$

$$\Rightarrow \frac{N_s}{V} = \text{const}$$

$$\Rightarrow \frac{f_s}{V} = \text{const.}$$